

devices that you want to monitor or control. Greengrass core act as the central hub. It performs the local execution of Lambda functions, transfers messages between devices, enables device shadows etc. A device should run Linux and should support ARM for hosting Greengrass.

Greengrass core can work with other devices as well. Especially, those that have Amazon FreeRTOS or IoT Device SDK. These devices can trigger the execution of AWS Lambda function locally even if IoT-core is unable to connect with the cloud.

Greengrass ML inference:

Now it has become possible to run Machine Learning locally with Greengrass Machine Learning inference capability. Let's say you are running a hotel, and want to locate your VIP customer and give them special treatment. What will you do? With Greengrass ML you can store a facial recognition model and run it on in-store cameras. When your camera identifies a VIP guest, it will trigger a lambda function. You can then inform your staff to tend to the VIP guest or display a greeting at the door. You need not to send huge amounts of data to the IoT cloud to achieve this. It can be done locally.

Benefits of AWS Greengrass:

- You can have real-time response as the devices act on data locally. Should the need arise, they use advanced AWS features as well.
- It allows your devices to operate even when there is no internet connectivity.
- Your data is always secure with AWS Greengrass. It encrypts data for cloud communication and local device to device communication. It uses mutual device authentication and authorisation at every point of network.
- AWS Greengrass uses the same Lambda programming you are familiar with. Additionally, it gives you the provision of running that Lambda function locally.
- It is one of the most cost effective solutions for IoT. Greengrass performs some action on the data that your device collects. So, all the data need not to be transmitted to the application in cloud. This significantly lowers the cost and increases the quality of the data.

Some Use cases of Greengrass:

- We have already seen in the previous section that you can use Greengrass in hospitality and retail industry for giving better treatment to your special guests.